AMIKACIN

Nephrotoxic & ototoxic

Monitor levels

CHEMICAL NAME (2S)-4-amino-N-[(1R,2S,3S,4R,5S)-5-amino-2-[(2S,3R,4S,5S,6R)-4-amino-3,5-dihydroxy-6-(hydroxymethyl)oxan-2-yl]oxy-4-[(2R,3R,4S,5R,6R)-6-(aminomethyl)-3,4,5-trihydroxyoxan-2-yl]oxy-3-hydroxycyclohexyl]-2-hydroxybutanamide (sulfate)

CLASS Semi-synthetic aminoglycoside antibiotic

TRADE NAMES Amikin · Amiklin · Arikayce (inhaled liposomal)

THERAPEUTIC USES / INDICATIONS

- Serious Gram-negative infections resistant to gentamicin (incl. *Pseudomonas*, *Acinetobacter*).
- Hospital-acquired sepsis, febrile neutropenia (combination).
- Multidrug-resistant tuberculosis; non-tuberculous mycobacteria.
- Neonatal sepsis in resistant settings.

ADMINISTRATION

- **IV** infusion over 30–60 min; **IM**.
- Do not mix with β-lactams in same line.
- Monitor peak/trough levels per protocol.

DOSAGES

- **Adult once daily:** 15 mg/kg IV once daily (max 1.5 g/day).
- **Multiple daily:** 7.5 mg/kg every 12 h.
- **Children:** 15–20 mg/kg once daily.
- **Neonates:** 15 mg/kg every 24–48 h by age.
- Extend interval in renal impairment; avoid trough >10 mg/L and peak >35 mg/L (conventional dosing; local targets for once-daily).

PHARMACOKINETICS

- **Onset:** IV immediate.
- Not metabolised; poor CSF penetration.
- Excreted unchanged in urine. **t½:** 2–3 h (prolonged in renal failure).

PHARMACODYNAMICS

- Binds **30S ribosome** → bactericidal protein-synthesis inhibition.
- Side chain protects against many aminoglycoside-modifying enzymes → active when gentamicin resistant.
- Concentration-dependent killing.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Injection pain.
- Nausea, rash.
- Headache.

ADVERSE EFFECTS (SERIOUS)

- Nephrotoxicity.
- Ototoxicity (mainly cochlear — hearing loss).
- Neuromuscular blockade, respiratory paralysis.
- Hypomagnesaemia, hypokalaemia.

CONTRAINDICATIONS & CAUTIONS

- Myasthenia gravis.
- Aminoglycoside hypersensitivity.
- **Caution:** renal impairment, elderly, pregnancy, hearing impairment.

KEY INTERACTIONS

- **Loop diuretics:** ↑ ototoxicity.
- **Vancomycin, amphotericin, ciclosporin, cisplatin:** ↑ nephrotoxicity.
- **Neuromuscular blockers:** enhanced block.

NURSING MANAGEMENT

- Calculate dose on correct weight; check renal function.
- Accurate timing of doses and levels.
- Daily creatinine, urine output; audiometry in long courses (MDR-TB).
- Ask about tinnitus, hearing loss, dizziness.
- Maintain hydration.

PATIENT EDUCATION

- Report hearing changes, ringing in ears, dizziness or less urine.
- Long TB courses need hearing tests.

KEY POINTS

- Aminoglycoside for gentamicin-resistant Gram-negatives.
- Hearing loss and kidney damage; monitor levels.
- 15 mg/kg once daily.
- Used in MDR-TB.

SOURCES CHECKED **[166]** Amikacin 250 mg/ml Injection (UK SmPC)

Full references p. 432–457